

Walk-Unbiased Doubly Robust Graph Propagation: Provably Fair, Transparent, Accountable and Private Recommendation from Exposure-Biased Logs

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Abstract

Graph-based recommenders propagate preferences along multi-hop walks of the logged user-item graph. This graph is filtered by the platform’s own exposure policy, so propagation amplifies exposure bias. Recent debiasing methods reweight the edges of this graph by inverse propensities or impute missing edges with doubly robust (DR) estimates. We show that these corrections are still biased once propagation spans more than one hop. A walk that revisits an edge squares that edge’s estimate, and the squared inverse propensity inflates exactly the rarely exposed edges that debiasing is meant to recover. We derive the exact bias and a lower bound that grows as the inverse of the propensity clip. Because outcomes are binary, the fix is simple: every repeated edge is replaced by its unbiased idempotent counterpart. The resulting operator, *DRUP* (Doubly Robust, walk-Unbiased Propagation), is exactly unbiased and edge-wise doubly robust for propagation over the counterfactual full-exposure graph, and it costs no more than the uncorrected operator. The same structure gives DRUP provable properties beyond accuracy: (i) variance, clipping-bias and concentration bounds; (ii) exposure invariance, a causal item-fairness guarantee that the expected score does not depend on interventions on an item’s exposure, whereas logged-graph propagation amplifies exposure and uncorrected DR propagation over-corrects it; (iii) scores that are affine in each user’s own interactions, which yields exact Shapley attributions and provably minimal counterfactual explanations; (iv) closed-form certificates against injected fake profiles; and (v) a single public item operator whose Gaussian release makes the whole recommender jointly differentially private. On Coat and KuaiRec, whose test sets are random or fully observed, DRUP needs no training. It is accuracy-competitive with trained MF, LightGCN, NAVIP and DR baselines (best on KuaiRec, tied with the best trained model on Coat), and every guarantee is verified empirically.

Keywords: causal recommendation; graph neural networks; doubly robust estimation; exposure bias; fairness; explainability; certified robustness; differential privacy.

1 Introduction

Graph-based recommender systems (GBRs) represent users and items as nodes and logged interactions as edges. They score a user-item pair by aggregating signals along multi-hop walks [4, 14]. The logged graph, however, is not the preference graph. An edge exists only if the platform *exposed* the item and the user then responded. A recent survey of causal learning in GBRs [17] traces four open challenges to this gap: out-of-distribution generalisation, dynamic preferences, fairness and interpretability. All four arise because message passing propagates correlations of the observational graph rather than causal preference structure. The same survey lists the lack of theoretical guarantees, scalability and privacy as open directions.

Two families of fixes exist. *Loss-level* debiasing (IPS, DR) reweights the training objective [12, 13, 15] but leaves the propagation operator biased. *Propagation-level* debiasing reweights edges by inverse propensities during neighbour aggregation [6, 8]. We show that the second family is itself biased for multi-hop propagation. The root cause is elementary. The edge estimate $W_e = O_e Y_e / p_e$ is unbiased for the potential outcome Y_e , but a walk that traverses e twice contributes W_e^2 , and $\mathbb{E}[W_e^2] = Y_e / p_e \neq Y_e$. Walks with repeated edges are not rare corner cases. They include every back-and-forth step $u \rightarrow j \rightarrow u$ that makes LightGCN self-reinforcing. Their bias is largest precisely where the propensity is smallest.

Contributions.

1. We identify the *repeated-walk bias* of inverse-propensity and DR graph propagation. We give its exact form and a lower bound that is unbounded as propensities vanish (Theorem 2).
2. We propose DRUP. Using $Y_e^2 = Y_e$, DRUP replaces each repeated edge by its idempotent unbiased counterpart. The resulting 3-hop operator is exactly unbiased and edge-wise doubly robust for propagation over the counterfactual full-exposure graph (Theorem 1). It has the cost of one item Gram matrix, needs no training, and factorises into a public item operator plus per-user local corrections.
3. We prove variance, clipping-bias and concentration bounds (Theorem 3).
4. We show that the same structure yields provable *FAT+P* properties: causal item fairness via exposure invariance (Theorem 4); exact attributions and optimal counterfactual explanations (Proposition 5, Theorem 6); certified robustness to injected profiles (Theorem 7); and joint differential privacy (Theorem 8).
5. We evaluate DRUP on Coat [13], whose test set is a random sample, and on KuaiRec [3], whose test matrix is fully observed. We compare it with trained and training-free baselines and verify each guarantee: by Monte Carlo, by a real-data intervention on exposure, by brute force, and against realised attacks.

2 Related work

Causal and debiased recommendation. Inverse propensity scoring [13] and doubly robust learning [15] are the standard tools for MNAR feedback. Causal embeddings disentangle interest from conformity [18]. The survey of [17] organises causal GBRs by challenge. None of these methods analyses the bias of the *propagation operator* beyond one hop.

Debiased message passing. NAVIP [8] applies inverse-propensity weights with Laplacian normalisation inside LightGCN aggregation. DPAA [6] learns popularity-aware aggregation weights. Both are unbiased for a single aggregation step at most. Theorem 2 shows that stacking such steps reintroduces a bias of order $1/p$.

Linear graph filters. LightGCN is a polynomial filter of the normalised adjacency, and closed-form filters are competitive with it [14]. DRUP is a closed-form filter on a debiased operator. Its linearity is what makes exact explanations, certificates and DP releases possible.

Explanations, robustness and privacy. GNNExplainer [16] and CF-GNNExplainer [9] approximate explanations by optimising on a surrogate. PORE [7] certifies recommenders against poisoning by ensembling. Differentially private recommenders release noisy item covariances [11]. DRUP gives exact explanations, closed-form certificates and a joint-DP guarantee [5] for one and the same scoring rule.

3 Problem setting

Users $u \in [m]$, items $i \in [n]$ and pairs $e = (u, i)$ carry a binary potential outcome Y_e and an exposure $O_e \sim \text{Bernoulli}(p_e)$. The log reveals O and $O \odot Y$.

Assumption 1 (Unconfoundedness). *Given the propensities, the exposures O_e are mutually independent and independent of Y .*

Assumption 2 (Fixed nuisances). *The propensity estimate $\hat{p}$, the imputation $\hat{Y} \in [0, 1]^{m \times n}$ and the edge weights C do not depend on O . This holds with cross-fitting or with nuisances estimated on a vetted historical log. We use $C_{ui} = d_u^{-\alpha} d_i^{-(1-\alpha)}$ with degrees taken from $\hat{Y}$.*

With $\tilde{Y} = C \odot Y$, the target is propagation over the full-exposure graph:

$$F^* = a\tilde{Y} + bT^*, \quad T^* = \tilde{Y}\tilde{Y}^\top \tilde{Y}. \quad (1)$$

This is the linear LightGCN / graph-filter score one would compute if every item had been shown to every user.

4 DRUP

Edge estimates. With clipped propensities $\bar{p} = \max(\hat{p}, \tau)$,

$$W_e^{\text{IPS}} = \frac{O_e Y_e}{\bar{p}_e}, \quad W_e^{\text{DR}} = \hat{Y}_e + \frac{O_e(Y_e - \hat{Y}_e)}{\bar{p}_e}. \quad (2)$$

Both have mean Y_e when $\bar{p} = p$.

Idempotent walk correction. Let $\tilde{W} = C \odot W$ and $\Delta = \tilde{W} \odot \tilde{W} - C \odot C \odot W$, with row sums r and column sums κ . The corrected 3-hop operator is

$$\hat{T} = \tilde{W}\tilde{W}^\top \tilde{W} - \tilde{W} \odot (r\mathbf{1}^\top - \Delta) - \tilde{W} \odot (\mathbf{1}\kappa^\top - \Delta) - (\tilde{W}^{\odot 3} - C^{\odot 3} \odot W). \quad (3)$$

The three corrections handle walks $u-j-v-i$ with $v = u$, with $j = i$, and with both. In each case every repeated factor $\tilde{W}_e^k$ is replaced by $C_e^k W_e$. The DRUP score is $s = a\tilde{W} + b\hat{T}$ with global constants a, b .

Public-operator form. Define the diagonal-corrected item Gram matrix $G = \tilde{W}^\top \tilde{W} - \text{diag}(\tilde{W}^\top \tilde{W}) + \text{diag}(\mathbf{1}^\top (C \odot C \odot W))$. Then

$$\hat{T}_u = \tilde{w}_u G - \tilde{w}_u \odot (r_u \mathbf{1}^\top - 2\delta_u) - (\tilde{w}_u^{\odot 3} - c_u^{\odot 3} \odot w_u), \quad (4)$$

where $\delta_u = \tilde{w}_u^{\odot 2} - c_u^{\odot 2} \odot w_u$. Equation (4) matches (3) to machine precision. G is the only object shared across users (Algorithm 1).

Algorithm 1 DRUP (training-free)

Require: $\log(O, O \odot Y)$; hyper-parameters $\tau, \alpha, \lambda, b/a$

- 1: fit propensities $\hat{p}$ (logistic exposure model) and imputation $\hat{Y}$ (IPS-weighted additive ridge)
 - 2: $\bar{p} \leftarrow \max(\hat{p}, \tau)$; $W \leftarrow \hat{Y} + O \odot (Y - \hat{Y})/\bar{p}$
 - 3: $C \leftarrow d_u^{-\alpha} d_i^{-(1-\alpha)}$ with d from $\hat{Y}$
 - 4: $G \leftarrow \tilde{W}^\top \tilde{W} - \text{diag}(\tilde{W}^\top \tilde{W}) + \text{diag}(\mathbf{1}^\top (C \odot \tilde{W}))$ ▷ public operator
 - 5: **for** each user u (locally) **do**
 - 6: $\hat{T}_u \leftarrow \text{Eq. (4)}$; $s_u \leftarrow a \tilde{w}_u + b \hat{T}_u$; rank unexposed items by s_u
 - 7: **end for**
-

Complexity. One $n \times n$ Gram product and $O(mn)$ reductions, i.e., the cost of the uncorrected operator. On KuaiRec (7,176 users, 10,728 items, 10.3M exposures) one configuration takes under a minute on four CPU cores.

5 Theory

Full proofs are in Appendix A. Let $\varepsilon_e = |Y_e - \hat{Y}_e|$.

Theorem 1 (Edge-wise double robustness). *Under Assumptions 1–2, let $\delta_e = (p_e/\bar{p}_e - 1)(Y_e - \hat{Y}_e)$. Then $\mathbb{E}[\hat{T}_{ui}]$ equals T_{ui}^* with every Y_e replaced by $Y_e + \delta_e$. In particular, if every edge has a correct propensity or a correct imputation, then $\mathbb{E}[s] = F^*$.*

Theorem 2 (Repeated-walk bias of uncorrected propagation). *With correct propensities and a candidate (u, i) with $O_{ui} = 0$, the uncorrected DR propagation $\tilde{W}\tilde{W}^\top \tilde{W}$ has conditional bias*

$$C_{ui}\hat{Y}_{ui}(R_u + K_i) - C_{ui}^3\hat{Y}_{ui}(1 - \hat{Y}_{ui}^2), \quad K_i = \sum_{v \neq u} C_{vi}^2 \frac{1 - p_{vi}}{p_{vi}} \varepsilon_{vi}^2,$$

and R_u is the analogous sum over the user’s own edges. The within-user ranking bias $C_{ui}\hat{Y}_{ui}K_i$ satisfies $K_i \geq (1/\max_v p_{vi} - 1) \sum_{v \neq u} C_{vi}^2 \varepsilon_{vi}^2$. It therefore diverges as the item’s propensities vanish, and it is $\Omega(1/\tau)$ under clipping. DRUP’s bias is zero.

Theorem 3 (Variance, clipping and concentration). *Let $\Lambda_e(u, i)$ be the C -weighted mass of the walks from u to i through edge e , and let $B = \max(1, 1/\tau)$. Then (a) $\text{Var}(W_e) \leq \varepsilon_e^2/\tau$; (b) $\text{Var}(\hat{T}_{ui}) \leq \frac{(1+1/\tau)^2}{\tau} \sum_e \varepsilon_e^2 \Lambda_e(u, i)^2$; (c) $|\mathbb{E}\hat{T}_{ui} - T_{ui}^*| \leq 3\bar{\delta}(1 + \bar{\delta})^2 \sum_{j,v} C_{uj}C_{vj}C_{vi}$ with $\bar{\delta} = \max_e(1 - p_e/\tau) + \varepsilon_e$; (d) $\Pr(|\hat{T}_{ui} - \mathbb{E}\hat{T}_{ui}| \geq t) \leq 2 \exp(-2t^2/\sum_e c_e^2)$ with $c_e = \varepsilon_e B^2 \Lambda_e(u, i)/\tau$.*

The variance scales with the imputation error ε^2 , not with Y^2 as for IPS, and the clip τ trades variance against a clipping bias that vanishes when $\hat{Y}$ is accurate.

Definition 1 (Exposure elasticity). For an intervention $do(p_{\cdot i} \leftarrow \pi p_{\cdot i})$ on item i ’s exposure mechanism, $\eta_i = \partial \log \mathbb{E}[s_{ui}]/\partial \log \pi$.

Theorem 4 (Exposure invariance: causal item fairness). *Under Assumptions 1–2 and $\pi p \geq \tau$, DRUP has $\eta_i = 0$, so items with equal potential outcomes receive equal expected scores whatever their exposure history. Logged-graph propagation has $\eta_i \geq 1$ (popularity amplification). Uncorrected DR propagation has $\eta_i < 0$ whenever $\hat{Y}_{ui}K_i > 0$ (over-correction).*

Proposition 5 (Affine in the user’s own data). *Fix u and a candidate i with $O_{ui} = 0$, and let $G^{(-u)}$ be G without u ’s own contribution. Then $s_{ui} = \text{const} + \sum_{j \neq i} (c_{uj} G_{ji}^{(-u)} + c_{ui} w_{ui} c_{uj}^2) w_{uj}$. Hence un-logging interaction j changes s_{ui} by exactly $\phi_j = a_j(w_{uj} - \hat{Y}_{uj})$. Removal effects add up over any subset, the ϕ_j are the exact Shapley values, and completeness holds exactly.*

Theorem 6 (Optimal minimal counterfactual explanation). *For candidates i, k with $s_{ui} > s_{uk}$, sorting $g_j = \phi_j^{(i)} - \phi_j^{(k)}$ in decreasing order and removing interactions until the cumulative sum exceeds the margin yields a minimum-cardinality set whose removal makes k outrank i . If the positive g_j do not sum above the margin, no such set exists. The cost is $O(n + d_u \log d_u)$.*

Theorem 7 (Certified robustness to injected profiles). *With frozen nuisances, one injected user v changes u ’s 3-hop score of item k by $\Delta_k = x_k(D + a_k c_{vk}) - a_k x_k^2$, where $a = \tilde{w}_u$, $x = c_v \odot w_v$ and $D = \langle a, x \rangle$. If v logs at most L interactions with arbitrary items, labels and propensities $\geq \tau$, then D and x_k lie in closed-form intervals. Δ_k is linear in D and quadratic in x_k , so its extremes $[\underline{\Delta}_k, \overline{\Delta}_k]$ over this box are exact. Item t provably stays out of u ’s top- K against any F such users if $s_{ut} + F \overline{\Delta}_t$ is below the $(K+1)$ -th largest value of $s_{uk} + F \underline{\Delta}_k$. With unclipped IPS, the influence of a single interaction is unbounded.*

Theorem 8 (Joint differential privacy). *Scale each user row so that $\|\tilde{w}_v\|_2 \leq R$ and $\|c_v^{\odot 2} \odot w_v\|_2 \leq R^2$. Releasing G plus symmetric Gaussian noise with $\sigma = 2R^2 \sqrt{2 \ln(1.25/\delta)}/\epsilon$ is (ϵ, δ) -DP under adding or removing one user. Scoring by (4) uses only the release and the user’s own row, so the recommender is (ϵ, δ) -jointly DP [5]. Post-processing such as low-rank denoising is free.*

6 Experiments

6.1 Setup

Data. *Coat* [13] has 290 users and 300 items. Training consists of 6,960 self-selected (MNAR) ratings. Testing uses 16 ratings per user on random items (MAR), and we rank only items the user never rated. The dataset ships propensities; we also fit our own exposure model, with the same conclusions. *KuaiRec* [3] provides the big matrix (7,176 users, 10,728 items, 10.3M logged views) as the MNAR log and a fully observed small matrix (1,411 users, 3,327 items, 99.6% dense) as the test set. We rank all never-logged small-matrix items. Positives are ratings ≥ 4 (Coat) and watch ratio ≥ 2 (KuaiRec).

Protocol. Hyper-parameters are chosen on a validation part of the unbiased data: 30% of each user’s test items for Coat, 30% of the test users for KuaiRec. We report the held-out remainder, averaged over 10 (Coat) or 5 (KuaiRec) random splits. Trained baselines are popularity, imputation only, and MF / IPS-MF / DR-MF / LightGCN / NAVIP-LightGCN / DR-LightGCN with 64-dimensional embeddings, pointwise losses and early stopping on the validation part. On Coat they use 10 splits and a grid over learning rate and weight decay. On KuaiRec, because of our CPU-only budget, they use 3 splits, one learning rate (10^{-2}), one weight decay (10^{-4}) and at most 15 epochs, so their KuaiRec numbers should be read as lower bounds. The training-free propagation family shares one scoring form and differs only in its operator: the logged graph (linear LightGCN), IPS adjacency (NAVIP-style), IPS with the walk correction (WC), DR adjacency, and DRUP.

6.2 Monte-Carlo verification

Table 1 uses a synthetic MNAR model with known propensities, 20,000 replications for bias and 3,000 for elasticity. The uncorrected IPS and DR operators have relative biases of 2.4 and 1.9.

Table 1: Monte-Carlo check of Theorems 1, 2 and 4.

Estimator	rel. —bias—	frac. $ z > 3$	rel. RMSE	exposure elasticity η
Obs	0.992	1.0000	1.02	+1.777
IPS	2.403	0.4042	21.01	+0.007
IPS+WC	0.028	0.0025	4.93	+0.007
DR	1.903	0.6292	20.94	-0.187
DRUP	0.020	0.0008	3.97	+0.005

Table 2: Accuracy on unbiased test data (mean $\pm$ std over splits). Best in bold, second underlined.

Method	metrics (Coat: nDCG@5, Recall@5, nDCG@10; KuaiRec: nDCG@20, Recall@20, nDCG@50)		
<i>Coat</i>			
Popularity	0.5367±0.0095	0.6441±0.0093	0.6599±0.0086
Imputation only	0.5662±0.0095	0.6736±0.0120	0.6787±0.0071
MF	0.4619±0.0224	0.5666±0.0213	0.6056±0.0178
IPS-MF	0.4569±0.0234	0.5579±0.0241	0.6049±0.0178
DR-MF	0.5357±0.0141	0.6473±0.0162	0.6575±0.0113
LightGCN	0.5077±0.0173	0.6036±0.0216	0.6403±0.0109
NAVIP-LightGCN	0.4994±0.0199	0.5979±0.0246	0.6329±0.0129
DR-LightGCN	0.5560±0.0241	0.6565±0.0254	0.6725±0.0165
Linear LightGCN (logged graph)	0.5571±0.0100	0.6461±0.0094	<u>0.6745±0.0076</u>
IPS adjacency (NAVIP-style)	0.5438±0.0186	0.6403±0.0152	<u>0.6641±0.0138</u>
IPS adjacency + WC	0.5438±0.0186	0.6403±0.0152	0.6641±0.0138
DR adjacency	<u>0.5586±0.0124</u>	<u>0.6728±0.0143</u>	0.6737±0.0088
DRUP (ours)	0.5526±0.0134	0.6637±0.0157	0.6720±0.0108
<i>KuaiRec</i>			
Popularity	0.5766±0.0023	0.1656±0.0016	0.4610±0.0019
Imputation only	0.6215±0.0014	0.1867±0.0020	0.5112±0.0013
MF	0.6253±0.0065	0.1925±0.0018	0.5080±0.0048
IPS-MF	0.6232±0.0051	0.1914±0.0021	0.5019±0.0049
DR-MF	0.6098±0.0068	0.1843±0.0034	0.4953±0.0034
LightGCN	0.6286±0.0006	0.1930±0.0013	0.5112±0.0018
NAVIP-LightGCN	0.6253±0.0031	0.1917±0.0014	0.5123±0.0016
DR-LightGCN	0.6158±0.0024	0.1879±0.0019	0.5032±0.0019
Linear LightGCN (logged graph)	0.6036±0.0028	0.1815±0.0016	0.4985±0.0024
IPS adjacency (NAVIP-style)	0.5954±0.0022	0.1829±0.0015	0.4892±0.0017
IPS adjacency + WC	0.5954±0.0022	0.1829±0.0015	0.4892±0.0017
DR adjacency	<u>0.6318±0.0017</u>	0.1927±0.0023	<u>0.5144±0.0019</u>
DRUP (ours)	0.6326±0.0018	<u>0.1929±0.0025</u>	0.5147±0.0015

With the walk correction, bias disappears: the fraction of entries with $|z| > 3$ matches the 0.27% expected by chance, and RMSE drops about fivefold. Elasticity follows Theorem 4: +1.78 for the logged graph, -0.19 for DR adjacency, and ≈ 0 for DRUP.

6.3 Accuracy

Table 2 reports accuracy and Table 3 the paired t -tests over shared splits. DRUP needs no training. On KuaiRec it has the highest nDCG@20 (0.6326). It is significantly better than the imputation model, logged-graph and IPS propagation, DR-MF and DR-LightGCN, and it is ahead of LightGCN, MF and NAVIP by margins that are not significant with three splits. On Coat it significantly outperforms MF, IPS-MF, DR-MF, LightGCN and NAVIP and ties with DR-LightGCN. It is, however, significantly *below* the imputation-only model (-0.014) and the uncorrected DR adjacency (-0.006). With 290 users and 16 random ratings each, the additive imputation already captures most of the recoverable signal. The walk correction therefore does not buy top- K accuracy. It

Table 3: Paired t -tests of DRUP against each baseline (nDCG@5 on Coat, nDCG@20 on KuaiRec).

Data	baseline	splits	mean diff. (DRUP – baseline)	p (paired t)
Coat	Pop	10	+0.0159	0.0021
Coat	Impute	10	-0.0136	0.000517
Coat	Obs	10	-0.0044	0.356
Coat	IPS	10	+0.0088	0.249
Coat	IPS+WC	10	+0.0088	0.249
Coat	DR	10	-0.0060	0.00272
Coat	MF	10	+0.0908	8.78e-08
Coat	IPS-MF	10	+0.0957	5.91e-07
Coat	DR-MF	10	+0.0170	0.000639
Coat	LightGCN	10	+0.0449	2.08e-06
Coat	NAVIP	10	+0.0533	3.06e-05
Coat	DR-LightGCN	10	-0.0033	0.635
KuaiRec	Pop	5	+0.0560	5.17e-07
KuaiRec	Impute	5	+0.0111	4.64e-06
KuaiRec	Obs	5	+0.0289	2.57e-05
KuaiRec	IPS	5	+0.0372	3.23e-06
KuaiRec	IPS+WC	5	+0.0372	3.23e-06
KuaiRec	DR	5	+0.0008	0.0758
KuaiRec	MF	3	+0.0066	0.197
KuaiRec	IPS-MF	3	+0.0086	0.0705
KuaiRec	DR-MF	3	+0.0221	0.0262
KuaiRec	LightGCN	3	+0.0033	0.0822
KuaiRec	NAVIP	3	+0.0066	0.154
KuaiRec	DR-LightGCN	3	+0.0161	0.00101

buys unbiasedness, exposure invariance and certifiability. As predicted by Theorem 2, IPS+WC coincides with IPS on unexposed candidates at three hops, because the factor $W_{ui} = 0$ hides the IPS bias. We do not claim a large accuracy gain. The claim is that DRUP is accuracy-competitive with trained models while adding the guarantees of Section 5.

6.4 Fairness

Table 4 tests Theorem 4 on real data. We thin the logged exposures of a random half of the items with probability $1/2$, which is an intervention with a known propensity change, and re-run each method. Logged-graph propagation demotes the thinned items strongly: their mean within-user rank shifts by -0.134 on KuaiRec and -0.058 on Coat. In other words, it punishes under-exposure. Clipped IPS demotes them less on KuaiRec (-0.029) and as much on Coat (-0.059), where clipping at $\tau = 0.2$ makes it almost identical to the logged graph. DRUP’s shift is $+0.0005 \pm 0.0008$ on KuaiRec and $+0.0006 \pm 0.011$ on Coat, i.e., zero within noise, as Theorem 4 predicts. The uncorrected DR adjacency is also close to zero on these data. Its over-correction ($\eta = -0.19$ in Table 1) scales with $(1 - p)/p \cdot \varepsilon^2$ (Theorem 2), and the validation-selected clip and an accurate imputation keep it small here. The exposure-conditional bias ECB (KuaiRec, where item quality is known from the fully observed matrix) drops from 0.830 (logged graph) and 0.571 (IPS) to 0.418 (DRUP). The nDCG gap between men and women on Coat shrinks from 0.112 to 0.079 . On KuaiRec, the gap between active and inactive users is small for every method (≤ 0.013).

Limitation. Exposure invariance is a causal guarantee, not a distributional one. When true

Table 4: Fairness. ECB is the partial correlation between an item’s mean rank and its log exposure, controlling for its true like rate. The user gap is the nDCG gap between user groups (gender on Coat, activity on KuaiRec). The last column is the rank shift of items whose exposures were thinned by a real $do(p \leftarrow p/2)$; 0 means exposure invariance.

Data	Method	nDCG	ECB↓	PRU↓	Gini↓	user gap↓	rank shift under $do(p/2)$
Coat	Obs	0.4917	0.769	0.394	0.340	0.1123	-0.0585±0.0073
Coat	IPS	0.4864	0.765	0.392	0.337	0.1189	-0.0594±0.0066
Coat	DR	0.4770	0.744	0.763	0.655	0.0795	-0.0027±0.0122
Coat	DRUP	0.4803	0.739	0.758	0.655	0.0793	+0.0006±0.0110
KuaiRec	Obs	0.6026	0.830	0.845	0.991	0.0080	-0.1342±0.0009
KuaiRec	IPS	0.5950	0.571	0.756	0.983	0.0023	-0.0290±0.0011
KuaiRec	DR	0.6324	0.418	0.831	0.994	0.0116	+0.0006±0.0007
KuaiRec	DRUP	0.6334	0.418	0.831	0.994	0.0129	+0.0005±0.0008

Table 5: Best nDCG reachable by each operator under a cap on the Gini coefficient of top- K exposure (over the whole hyper-parameter grid).

Data	Method	Gini cap			
		≤ 0.4	≤ 0.5	≤ 0.6	none
Coat	Obs	0.4917	0.4917	0.4917	0.4917
Coat	IPS	0.4864	0.4864	0.4864	0.4864
Coat	DR	0.4090	0.4487	0.4733	0.4853
Coat	DRUP	0.3786	0.4416	0.4848	0.4902
		≤ 0.97	≤ 0.98	≤ 0.99	none
KuaiRec	Obs	0.3543	0.3543	0.3543	0.6026
KuaiRec	IPS	0.5398	0.5745	0.5950	0.5950
KuaiRec	DR	–	0.3394	0.4633	0.6324
KuaiRec	DRUP	–	–	–	0.6328

quality is concentrated, DRUP’s top- K lists are concentrated too, with higher Gini and PRU than the logged graph. The accuracy–concentration frontier (Table 5) makes this trade-off explicit. On Coat, DRUP reaches only 0.379 nDCG@5 under a Gini cap of 0.4, while the logged graph reaches 0.492. On KuaiRec, IPS keeps 0.575 at Gini ≤ 0.98 , while every DRUP configuration in our grid exceeds 0.99. Combining DRUP with an exposure-constrained re-ranker is left for future work.

6.5 Transparency

Attributions recomputed against full re-propagation are exact up to floating-point precision: 1.3×10^{-15} on Coat (float64) and 3×10^{-6} on KuaiRec (float32), for completeness as well as for additivity over random subsets (Table 6). Removing the interactions with the largest attributions moves the top item further down than random or similarity-based removal. This holds for every removal budget on KuaiRec and for up to five removals on Coat, and it shows that the attributions are faithful. The absolute rank changes are small, because top recommendations rest largely on population-level evidence. Theorem 6 also answers a question approximate explainers cannot: whether any counterfactual explanation exists at all. On Coat, 82% of the top recommendations cannot be overturned by removing any subset of the user’s own interactions. On KuaiRec the figure is 45%, and the minimal explanation has a median size of 15.5 interactions. These recommendations are driven by

Table 6: Exact explanations (Prop. 5, Thm. 6) on 200 random test users.

Data	max rel. error	mean min. CF size	median	users without CF
Coat	1.3e-15	2.36	2	0.82
KuaiRec	3.3e-06	24.40	16	0.45

Table 7: Injection attack: F fake users log a low-popularity target item and the most popular items. Left: realised hit rate of the target in real users’ top- K . Right: fraction of real users for whom Theorem 7 certifies that the target cannot enter the top- K against *any* F profiles of that size. No certificate was violated.

<i>Coat: $K=5$, fake profile size 21</i>																
	hit rate of target in top- K ↓							certified fraction ↑								
F	0	1	2	5	10	20	50	0	1	2	5	10	20	50		
Obs	0.004	0.004	0.004	0.009	0.108	0.444	0.661	0.996	0.000	0.000	0.000	0.000	0.000	0.000		
IPS	0.004	0.004	0.004	0.020	0.201	0.502	0.651	0.995	0.000	0.000	0.000	0.000	0.000	0.000		
DR	0.000	0.000	0.000	0.000	0.000	0.000	0.655	1.000	0.000	0.000	0.000	0.000	0.000	0.000		
DRUP	0.000	0.000	0.000	0.000	0.000	0.382	0.973	1.000	1.000	0.996	0.211	0.000	0.000	0.000		
<i>KuaiRec: $K=20$, fake profile size 51</i>																
	hit rate of target in top- K ↓								certified fraction ↑							
F	0	1	2	5	20	50	100	200	0	1	2	5	20	50	100	200
Obs	0.002	0.002	0.002	0.002	0.002	0.002	0.027	0.288	0.998	0.000	0.000	0.000	0.000	0.000	0.000	0.000
IPS	0.002	0.002	0.002	0.002	0.002	0.005	0.089	0.485	0.998	0.000	0.000	0.000	0.000	0.000	0.000	0.000
DR	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.135	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
DRUP	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.132	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

population-level evidence, and DRUP certifies that they are.

6.6 Accountability: certified robustness

Table 7 reports the attack. Empirically, the DR-based operators are the hardest to manipulate. On KuaiRec, 100 fake users (1.4% of the user base) never place the target in any real user’s top-20 under DR or DRUP, while they already do so for 2.7% (logged graph) and 8.9% (IPS) of users. At 200 fake users the rates are 13% for DR and DRUP, against 29% and 49%. On Coat, DRUP resists up to 10 fake users, while the logged graph and IPS are breached at 5–10.

The certificates of Theorem 7 were never violated, and brute-force enumeration confirms the bounds in float32 and float64. They are informative when catalogs are small. On Coat, DRUP is the only operator with non-trivial certificates: 100% of users are certified against one 21-interaction fake profile, 99.6% against two and 21% against five. The reason is that degrees taken from the imputation make the fake user’s normalisation known, whereas degrees taken from W are attacker-controlled and void the certificate. On KuaiRec the worst-case certificate is vacuous even for $F = 1$. With 10,728 items, a profile of 51 interactions tailored to a single victim can in the worst case shift scores by more than the gap between a low-popularity target and the top-20 boundary. The realised attacks stay far below this worst case. Tighter certificates for large catalogs, for instance under bounded profile norms, are an open problem. At intermediate budgets the uncorrected DR adjacency with attacker-controlled degrees is sometimes harder to push than DRUP, because the fake user’s large DR weights inflate its own degree. That protection cannot be certified, because an attacker can design a profile that avoids it.

Table 8: nDCG of jointly DP DRUP ($\delta = 10^{-5}$). The denoising rank is chosen on validation data.

Data	$\epsilon=0.5$	1	2	4	8	16	∞ (clipped)	non-private / Pop
Coat	0.4094	0.4118	0.4182	0.4651	0.5068	0.5301	0.5398	0.5326 / 0.5226
KuaiRec	0.0536	0.0826	0.1473	0.5569	0.6127	0.6290	0.6346	0.6325 / 0.5766

6.7 Privacy

Table 8 shows the privacy–utility curve. On KuaiRec (7,176 users), jointly DP DRUP reaches nDCG@20 of 0.613 at $\epsilon = 8$ and 0.629 at $\epsilon = 16$. The non-private value is 0.633, trained LightGCN reaches 0.629 and popularity 0.577. On Coat, with only 290 users, the same guarantee costs considerably more accuracy. Useful privacy needs a large user base, as expected for user-level DP.

7 Discussion and limitations

DRUP relies on unconfoundedness and on nuisances that are fixed or cross-fitted, the standard assumptions of IPS/DR estimation. The exact correction is derived for up to three hops (and for the item Gram operator). Higher orders need the same idempotent rule, but the number of repetition patterns grows. The fairness guarantee concerns causal exposure invariance, not parity of outcomes. The privacy analysis treats the low-dimensional nuisances as public. Beyond accuracy, DRUP turns a graph recommender into an object whose bias, fairness, explanations, robustness and privacy can be stated and checked.

References

- [1] C. Dwork and A. Roth. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9(3–4):211–407, 2014.
- [2] B. Efron and C. Stein. The jackknife estimate of variance. *The Annals of Statistics*, 9(3): 586–596, 1981.
- [3] C. Gao, S. Li, W. Lei, J. Chen, B. Li, P. Jiang, X. He, J. Mao, and T.-S. Chua. KuaiRec: A fully-observed dataset and insights for evaluating recommender systems. In *Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM)*, pages 540–550, 2022.
- [4] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, and M. Wang. LightGCN: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 639–648, 2020.
- [5] J. Hsu, Z. Huang, A. Roth, T. Roughgarden, and Z. S. Wu. Private matchings and allocations. In *Proceedings of the 46th Annual ACM Symposium on Theory of Computing (STOC)*, pages 21–30, 2014.
- [6] M. A. Islam, A. S. Faruk, S. Medya, and E. Zheleva. Debiasing message passing to mitigate popularity bias in GNN-based collaborative filtering, 2026.

- [7] J. Jia, Y. Liu, Y. Hu, and N. Z. Gong. PORE: Provably robust recommender systems against data poisoning attacks. In *Proceedings of the 32nd USENIX Security Symposium*, pages 1703–1720, 2023.
- [8] M. Kim, J. Oh, J. Do, and S. Lee. Debiasing neighbor aggregation for graph neural network in recommender systems. In *Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM)*, 2022. doi: 10.1145/3511808.3557576.
- [9] A. Lucic, M. A. ter Hoeve, G. Tolomei, M. de Rijke, and F. Silvestri. CF-GNNExplainer: Counterfactual explanations for graph neural networks. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 4499–4511, 2022.
- [10] C. McDiarmid. On the method of bounded differences. In *Surveys in Combinatorics*, pages 148–188. Cambridge University Press, 1989.
- [11] F. McSherry and I. Mironov. Differentially private recommender systems: Building privacy into the Netflix prize contenders. In *Proceedings of the 15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 627–636, 2009.
- [12] Y. Saito, S. Yaginuma, Y. Nishino, H. Sakata, and K. Nakata. Unbiased recommender learning from missing-not-at-random implicit feedback. In *Proceedings of the 13th International Conference on Web Search and Data Mining (WSDM)*, pages 501–509, 2020.
- [13] T. Schnabel, A. Swaminathan, A. Singh, N. Chandak, and T. Joachims. Recommendations as treatments: Debiasing learning and evaluation. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, pages 1670–1679, 2016.
- [14] Y. Shen, Y. Wu, Y. Zhang, C. Shan, J. Zhang, K. B. Letaief, and D. Li. How powerful is graph convolution for recommendation? In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM)*, pages 1619–1629, 2021.
- [15] X. Wang, R. Zhang, Y. Sun, and J. Qi. Doubly robust joint learning for recommendation on data missing not at random. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 6638–6647, 2019.
- [16] R. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec. GNNExplainer: Generating explanations for graph neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [17] D. Yu, Q. Li, B. Soundararaj, C. Pettit, and G. Xu. A survey of causal learning in graph-based recommender systems. *ACM Computing Surveys*, 58(16):404:1–404:32, 2026. doi: 10.1145/3840748.
- [18] Y. Zheng, C. Gao, X. Li, X. He, Y. Li, and D. Jin. Disentangling user interest and conformity for recommendation with causal embedding. In *Proceedings of the Web Conference (WWW)*, pages 2980–2991, 2021.

A Proofs

Theorem 1. After the correction every walk contributes $\prod_{e \in \text{distinct}} C_e^{k_e} W_e$. The W_e are independent (Assumption 1, fixed nuisances), so $\mathbb{E} \prod W_e = \prod \mathbb{E} W_e$ with $\mathbb{E} W_e = \hat{Y}_e + p_e(Y_e - \hat{Y}_e) / \bar{p}_e = Y_e + \delta_e$. The target monomial is $\prod C_e^{k_e} Y_e^{k_e} = \prod C_e^{k_e} Y_e$ because $Y_e \in \{0, 1\}$. $\square$

Theorem 2. The naive operator contains, for $v = u$, $j \neq i$, the terms $\tilde{W}_{uj}^2 \tilde{W}_{ui}$; for $j = i$, $v \neq u$, the terms $\tilde{W}_{ui} \tilde{W}_{vi}^2$; and $\tilde{W}_{ui}^3$. Conditionally on $O_{ui} = 0$ we have $W_{ui} = \hat{Y}_{ui}$. By independence, $\mathbb{E}[W_e^2] = Y_e + \text{Var}(W_e)$ with $\text{Var}(W_e) = \frac{1-p_e}{p_e}(Y_e - \hat{Y}_e)^2$, while the corrected replacement has mean Y_e . Summing the excesses gives the formula. The lower bound uses $(1-p)/p \geq 1/\max p - 1$. $\square$

Theorem 3. (a) $\text{Var}(W_e) = p_e(1-p_e)\varepsilon_e^2/\bar{p}_e^2$. If $p_e \geq \tau$ this is at most $(1/\tau - 1)\varepsilon_e^2$; otherwise it is at most $p_e\varepsilon_e^2/\tau^2 < \varepsilon_e^2/\tau$. (b) $\hat{T}_{ui}$ is multilinear in independent variables and therefore affine in each W_e . The Efron–Stein inequality [2] gives $\text{Var}(\hat{T}) \leq \sum_e \mathbb{E}[(\partial_e \hat{T})^2] \text{Var}(W_e)$. The derivative $\partial_e \hat{T}$ is a sum, over the walks through e , of C -weights times at most two other independent W_f with $\|W_f\|_2^2 \leq 1 + 1/\tau$. Minkowski’s inequality then bounds $\|\partial_e \hat{T}\|_2 \leq (1 + 1/\tau)\Lambda_e$. (c) By Theorem 1, the bias is at most $\sum |\kappa_{\text{walk}}|[\prod(Y_f + |\delta_f|) - \prod Y_f] \leq ((1 + \bar{\delta})^3 - 1)M_{ui}$, and $|\delta_e| = (1 - p_e/\tau)_+ \varepsilon_e$. (d) Changing O_e moves W_e by at most ε_e/τ , and $|\partial_e \hat{T}| \leq B^2 \Lambda_e$. Apply McDiarmid’s inequality [10]. $\square$

Theorem 4. DRUP: by Theorem 1, $\mathbb{E}[s] = F^*$, which does not depend on π . Logged graph: $\mathbb{E}[O_e Y_e] = p_e Y_e$, so each 3-hop monomial containing $k \geq 1$ edges incident to i scales as π^k . Uncorrected DR: by Theorem 2, $\mathbb{E}[s] = F^* + C_{ui} \hat{Y}_{ui} K_i(\pi) + \dots$, and $K_i(\pi)$ is strictly decreasing in π . $\square$

Proposition 5 and Theorem 6. Substitute $G = G^{(-u)} + \tilde{w}_u^\top \tilde{w}_u - \text{diag}(\tilde{w}_u^2) + \text{diag}(c_u^2 w_u)$ into (4). The terms $\tilde{w}_{ui} \sum_j \tilde{w}_{uj}^2$ cancel against the local correction, leaving $\tilde{w}_{ui} \sum_j c_{uj}^2 w_{uj}$, and every remaining term involving w_{uj} ($j \neq i$) is linear. Additive games have Shapley values equal to their individual effects. For Theorem 6, the margin after removing S is $s_{ui} - s_{uk} - \sum_{j \in S} g_j$, so minimising $|S|$ under $\sum_S g_j > \text{margin}$ selects the largest g_j . $\square$

Theorem 7. Adding row v adds contrib_v to G and leaves u ’s local terms unchanged. Then $\Delta = \tilde{w}_u \text{contrib}_v = (a \cdot x)x - a \odot x^2 + a \odot c_v \odot x$. Logged DR entries satisfy $w \in [\hat{Y} - \hat{Y}/\tau, \hat{Y} + (1 - \hat{Y})/\tau]$ for any label and any $p \geq \tau$, and unlogged ones equal $\hat{Y}$. D is maximised (minimised) by assigning the L slots to the largest positive (negative) deviations. For fixed x_k the function is linear in D , so it attains its extremes at the ends of the D interval. For fixed D it is a quadratic in x_k , whose extremes over an interval lie at the ends or at the clamped vertex. Effects add over fake users. For unclipped IPS, $|W_e| = 1/p_e$ is unbounded. $\square$

Theorem 8. $\|\text{contrib}_v\|_F \leq \|\tilde{w}_v \tilde{w}_v^\top - \text{diag}(\tilde{w}_v^2)\|_F + \|c_v^2 w_v\|_2 \leq R^2 + R^2$, so the ℓ_2 -sensitivity of the upper triangle is at most $2R^2$, and the Gaussian mechanism [1] applies. Each user’s output is a function of the DP release and their own data, which gives joint DP by the billboard lemma [5]. $\square$